Newton’s Laws of Motion

Answer key

Conventions used throughout

• **Forces:** F_g = force of gravity (weight, $F_g = mg$), F_N = normal force, F_{air} = air resistance, F_{net} = net force (all the forces added together). “A on B” means the force that object A exerts on object B.

• **Free-body diagrams (FBDs):** the object is a dot, every force starts at the dot, and a longer arrow means a bigger force.

• **Signs in the working:** +y is up. **Directions in final answers:** [E], [W], [N], [S], [Up], [Down].

• $g = 9.81\ m/s^2$.

CONCEPTUAL 1

Examples of the three laws

1) Can you think of some real-life examples of when?

   a. Newton’s First Law is at play?

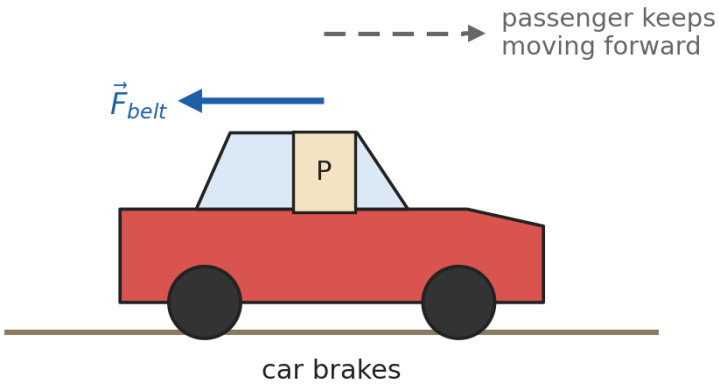
   b. Newton’s Second Law is at play?

   c. Newton’s Third Law is at play?

(a) Newton’s First Law is at play?

A car brakes hard. The passenger’s body keeps moving forward at the old speed (inertia) until the seat belt pulls back on it. No force, no change in motion.

Picture: first law

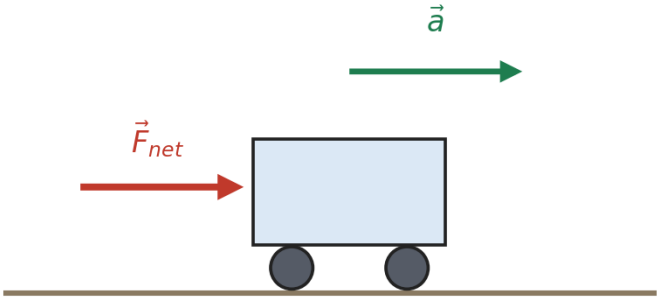


(b) Newton’s Second Law is at play?

Pushing a shopping cart. The harder you push (bigger net force), the faster it speeds up. A full cart (bigger mass) speeds up less for the same push.

Formula: $\vec{F}_{net} = m\vec{a} \quad \Rightarrow \quad a = \frac{F_{net}}{m}$

Picture: second law

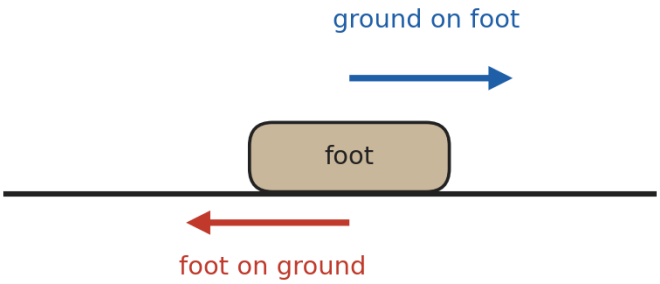


shopping cart

(c) Newton’s Third Law is at play?

Walking. Your foot pushes backward on the ground, and the ground pushes forward on your foot with an equal force. That forward push from the ground is what moves you.

Picture: third law



FINAL ANSWER (SAMPLE)

(a) A passenger keeps moving forward when a car brakes, until the seat belt stops them

(b) A harder push makes a shopping cart speed up faster

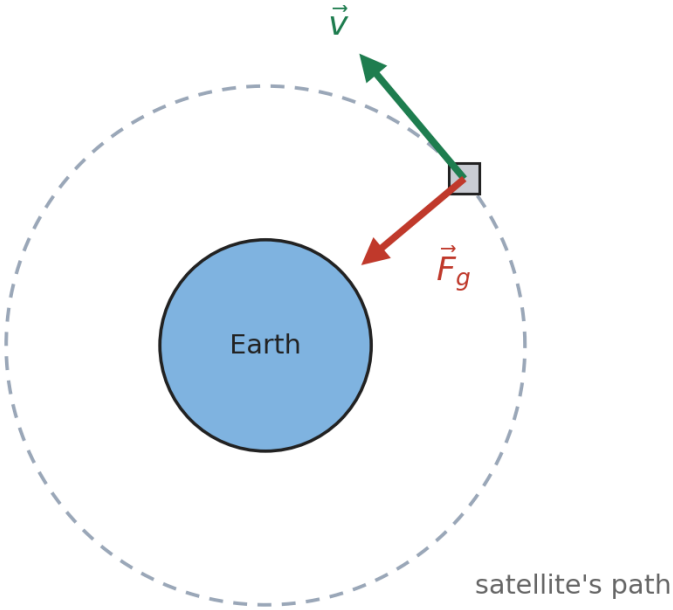
(c) Walking: foot pushes the ground back, ground pushes the foot forward

FIRST LAW 1

Satellite in orbit

1) True or False: Objects in orbit around Earth (like a satellite) must have a net force acting on them. Justify your answer.

Picture the situation



Step 1 – Is the velocity changing?

Velocity has a size and a direction. The satellite's speed may stay the same, but its direction changes all the time as it curves around Earth. A change in velocity is an acceleration.

Step 2 – Use the first law

By the first law, an object with no net force moves in a straight line at a constant speed. The satellite does not move in a straight line, so there must be a net force. Gravity from Earth, pulling toward Earth's centre, is that force.

FINAL ANSWER

True. The satellite's direction keeps changing, so a net force must act: gravity toward Earth.

FIRST LAW 2

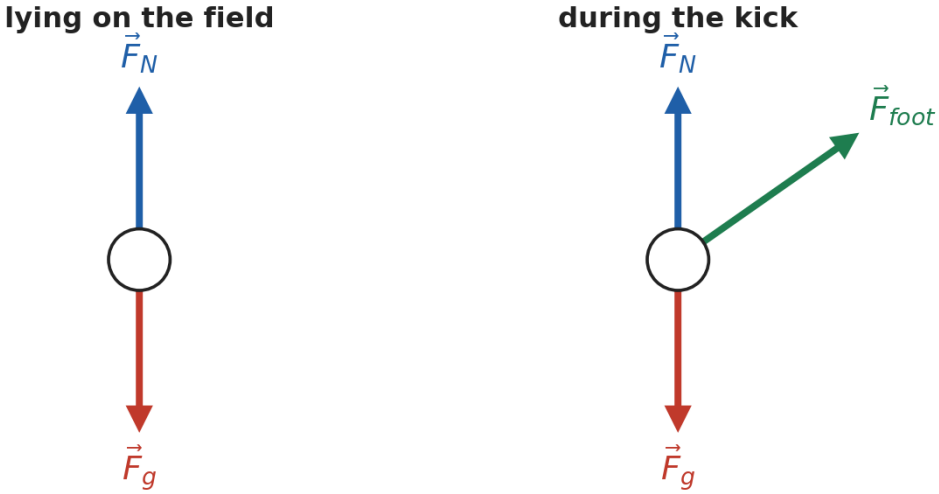
The soccer ball

2) Cristiano Ronaldo is racing against Neymar to get to the soccer ball that is just laying on the field. Cristiano gets there first and kicks the ball into the air.

a. Why is this a case of Newton's first law?

b. Is this an example of something remaining at rest or remaining in motion?

FBDs of the ball



(a) Why is this a case of Newton's first law?

While the ball lies on the field, F_N and F_g are equal and opposite, so the net force is zero and the ball stays still. It only starts moving when the kick adds an unbalanced force.

(b) Is this an example of something remaining at rest or remaining in motion?

Before the kick the ball is at rest, and it stays at rest until Cristiano's foot acts on it.

FINAL ANSWER

(a) The ball stays still until an unbalanced force (the kick) acts on it

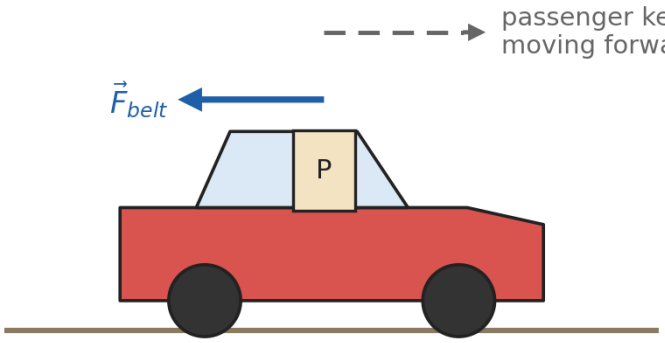
(b) Remaining at rest

FIRST LAW 3

Safety and inertia

3) Think about an example where Newton's First law plays a hand in safety within our everyday lives. What is happening in those scenarios that help keep us safe?

Picture the situation



car brakes

Seat belts. When a car stops suddenly, your body keeps moving forward because of inertia. The seat belt supplies the backward force that slows you down with the car, so you don't hit the dashboard or windshield. Airbags and car-seat headrests work the same way.

FINAL ANSWER (SAMPLE)

Seat belts: your body keeps moving forward when the car stops, and the belt applies the force that stops you safely

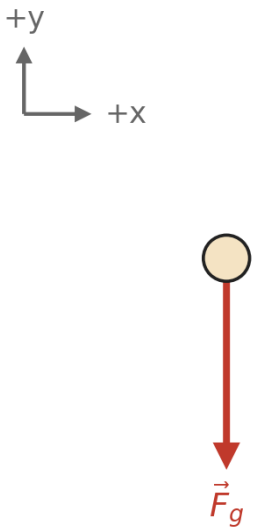
SECOND LAW 1

Free fall

A falling object has only one force acting on it – gravity (ignoring air resistance)

1) What is the acceleration of a freely falling object?

FBD of the falling object



Step 1 – Use the second law

Gravity is the only force, so it is the net force. Put $F_{net} = -mg$ (down is negative) into the second law:

Formula: $F_{net} = ma$

Put in $F_{net} = -mg$: $-mg = ma$

Divide both sides by m: $a = -g = -9.81\ \text{m/s}^2$

The mass cancels, so every object falls with the same acceleration.

FINAL ANSWER

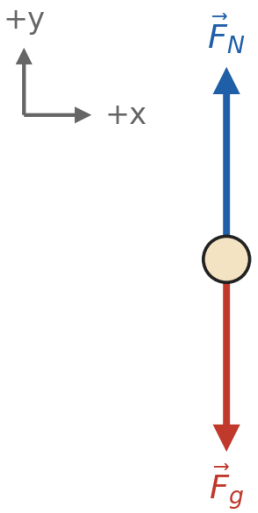
$\vec{a}$ = 9.81 m/s^2 [Down]

SECOND LAW 2

Standing still

2) If I am standing straight up (not moving just standing), what are the forces acting upon me? What is my NET force then??

FBD of the person



Step 1 – Find the net force

Two forces act: gravity pulls down and the floor pushes up (normal force). You are not accelerating, so a = 0:

Formula: $F_{net} = ma$

Put in $a = 0$: $F_{net} = m(0) = 0$ N

$F_{net} = F_N - F_g = 0 \quad \Rightarrow \quad F_N = F_g$

FINAL ANSWER

Forces: gravity (down) and the normal force from the floor (up)

Net force: $\vec{F}_{net}$ = 0 N

SECOND LAW 3

Air resistance during a fall

3) Recall that air resistance is proportional to speed squared. Imagine you drop an object off a tall building from rest. Draw a free body diagram of the object showing the forces of gravity and air resistance (it is important you draw the vectors of an appropriate length to correctly depict the situation)

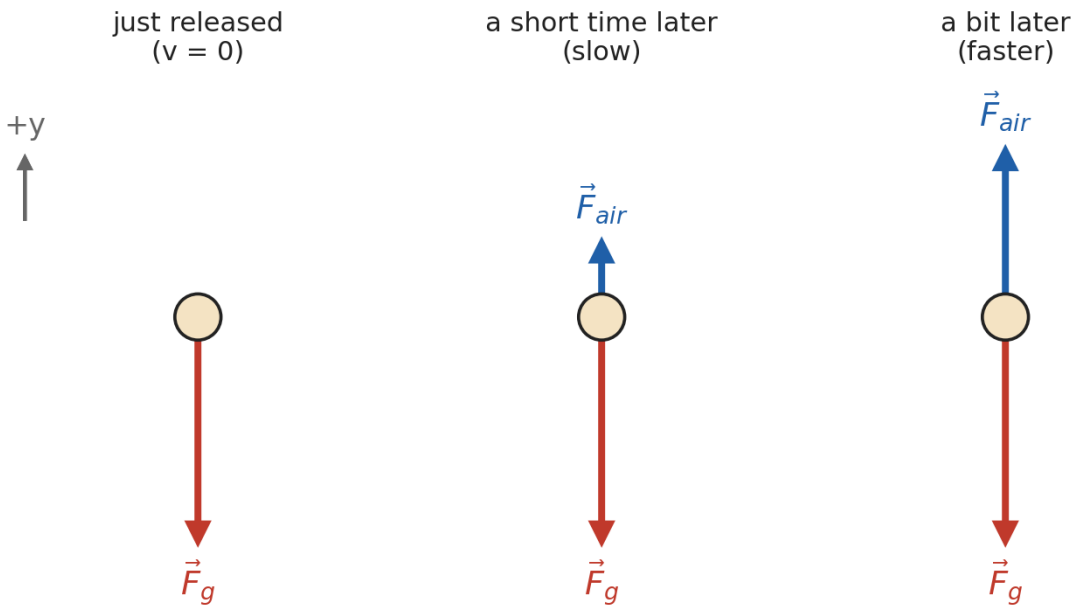
When it first leaves your hand

A short time after it leaves your hand when it is still travelling slowly

A bit later when it is travelling more quickly

Explain

FBDs at the three moments (F_g drawn the same length in all three)



Step 1 – Why the arrows change

$F_g = mg$ never changes, so its arrow stays the same length. Air resistance depends on speed ($F_{air} \propto v^2$), so it grows as the object speeds up.

Moment	Explain
First leaves your hand	v = 0, so $F_{air} = 0$. Only gravity acts, so a = 9.81 m/s^2 [Down].
Short time later (slow)	Small F_{air} up. F_g is still much bigger, so the net force is down and it keeps speeding up, but a little less quickly.
A bit later (faster)	Bigger F_{air}, close to F_g. The net force is small, so it speeds up only slowly.

FINAL ANSWER

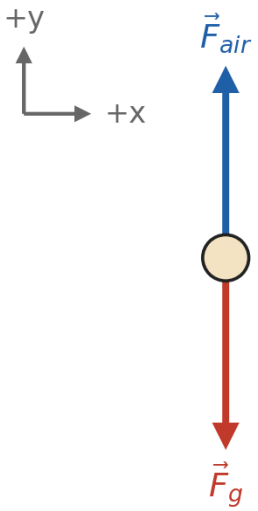
F_g stays the same; F_{air} grows with speed, so the net force and the acceleration shrink

SECOND LAW 4

Can air resistance exceed gravity?

4) Can the force of air resistance ever become larger than the force of gravity for a free-falling object? Explain (consider Newton's second law that discusses object that accelerate when there exits an unbalanced force).

FBD at terminal velocity



Step 1 – Follow the forces as it speeds up

F_{air} grows as the object speeds up. When $F_{air} = F_g$, the net force is zero:

Formula: $F_{net} = F_{air} - F_g$

Put in $F_{air} = F_g$: $F_{net} = 0 \quad \Rightarrow \quad a = 0$

With a = 0 the speed stops increasing (terminal velocity), so F_{air} stops growing. It never gets past F_g.

Step 2 – The one exception

If something makes the object move faster than its terminal velocity (for example, when a skydiver opens a parachute), F_{air} is briefly bigger than F_g. The net force is then up, so the object slows down until the two forces are equal again.

FINAL ANSWER

No, not for an object dropped from rest: air resistance grows only until it equals gravity (terminal velocity)

It is only bigger briefly if the object is moving faster than its terminal velocity, and then it slows down

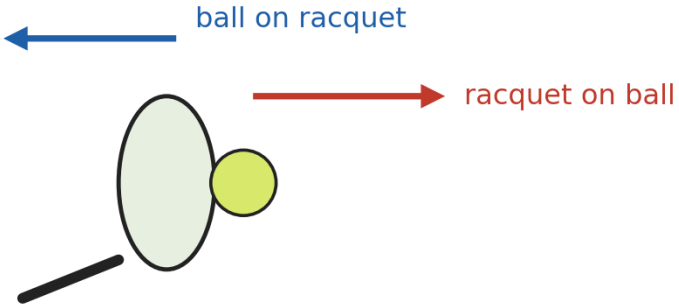
THIRD LAW

Tennis

Tennis a beautiful sport because we you can see Newton's laws all in action (and other reasons). During a serve, the mass of the tennis ball remains constant, but the more force you apply to the constant mass of the tennis ball onto the serve, the greater the acceleration will be. What law is this an example of?

Whenever a tennis player strikes a ball the force of the racquets striking the ball and the force of the ball striking the racquet both act upon each other in equal but opposite ways. What law is this an example of?

Picture the situation



Prompt 1 During a serve, the mass of the tennis ball remains constant, but the more force you apply to the constant mass of the tennis ball onto the serve, the greater the acceleration will be. What law is this an example of?

Force and acceleration are linked by the second law. With m fixed, a bigger F_{net} gives a bigger a:

Formula: $a = \frac{F_{net}}{m}$

Prompt 2 Whenever a tennis player strikes a ball the force of the racquets striking the ball and the force of the ball striking the racquet both act upon each other in equal but opposite ways. What law is this an example of?

Two objects push on each other with equal and opposite forces at the same time: that is the third law.

FINAL ANSWER

Prompt 1: Newton’s second law

Prompt 2: Newton’s third law

THIRD LAW 1

Action and reaction

1) For each of the following “action forces” state the reaction force (don’t forget to include the direction). Draw a mini picture of the scenario.

a. You are playing tennis with the wall. The ball exerts a force of 40 N [E] onto the wall.

b. A table exerts a normal force of 25 N [Up] on a book.

c. Your hand exerts a force of 90 N [N] on a basketball.

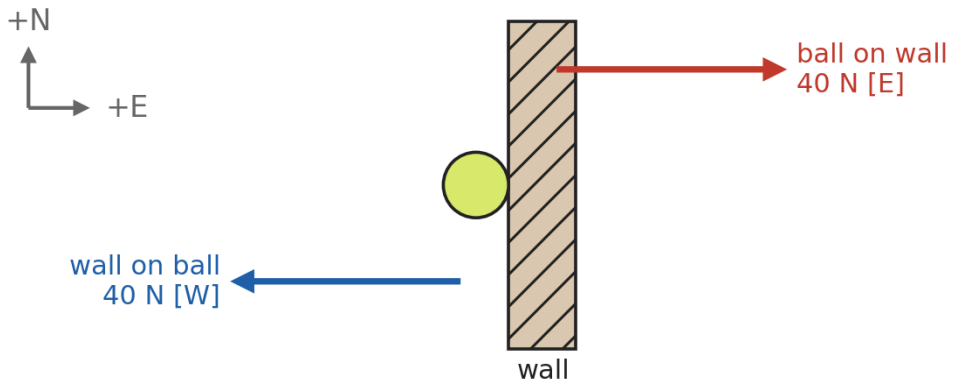
Rule for every part

The reaction force is the same size, in the opposite direction, and acts on the other object. Swap the two object names (“A on B” becomes “B on A”).

(a) You are playing tennis with the wall. The ball exerts a force of 40 N [E] onto the wall.

Action: ball on wall, 40 N [E]. Reaction: wall on ball, same size, opposite direction.

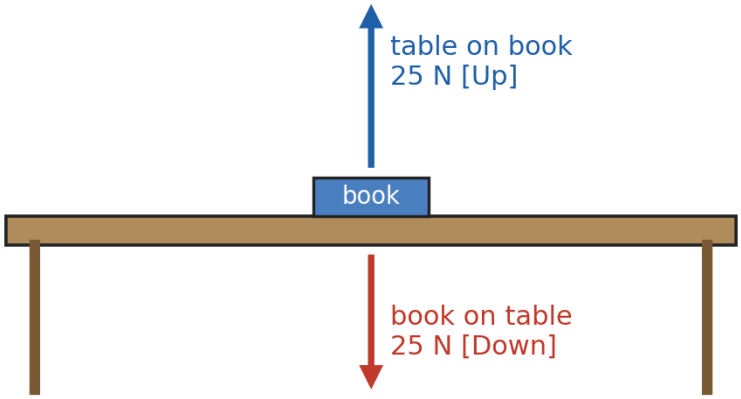
Mini picture



(b) A table exerts a normal force of 25 N [Up] on a book.

Action: table on book, 25 N [Up]. Reaction: book on table, same size, opposite direction.

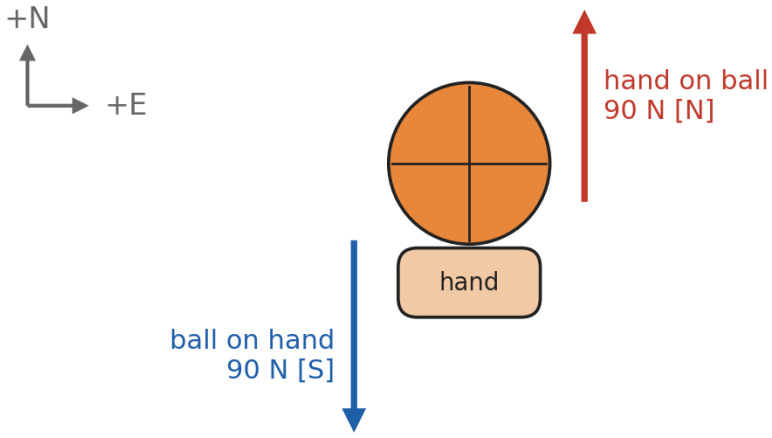
Mini picture



(c) Your hand exerts a force of 90 N [N] on a basketball.

Action: hand on ball, 90 N [N]. Reaction: ball on hand, same size, opposite direction.

Mini picture



FINAL ANSWER

(a) $\vec{F}_{\text{wall on ball}} = 40\ \text{N [W]}$

(b) $\vec{F}_{\text{book on table}} = 25\ \text{N [Down]}$

(c) $\vec{F}_{\text{ball on hand}} = 90\ \text{N [S]}$